WASP-2b

R-0095

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_171108` · created 2026-07-15T23:04:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458065.6641 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.047 +/- 0.045
Transit depth (Rp/Rs)^2	0.22 +/- 0.43 [%]
Orbital Inclination (inc)	89.81 +/- 2.88
Airmass coefficient 1 (a1)	0.99 +/- 0.6
Airmass coefficient 2 (a2)	0.05 +/- 0.12
Scatter in the residuals of the lightcurve fit is	58.82 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	5.5
Transit Duration (day)	0.082 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.047 ± 0.045 vs 0.1326 (deviation 1.9σ)
Duration fit vs published	0.082 d vs 0.0737 d (deviation 0.59σ)
Mid-time O–C	-10.8 min
Depth SNR	1.0
Residual scatter	58.82 %

Light curve

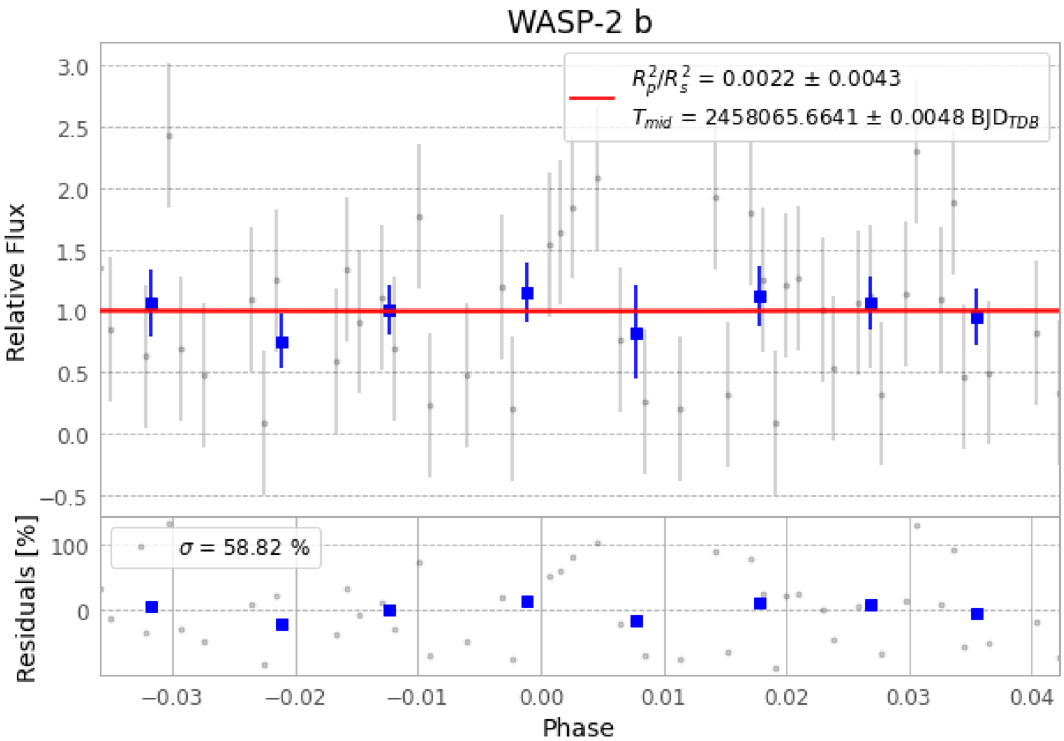

Figure 1 — FinalLightCurve_WASP-2 b_2017-11-07.png (SHA-256 9357b00413b99b26…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 99], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 401c2e6743d03167…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

82 FITS frames (54 MB), `WASP-2171108020343.FITS` → `WASP-2171108060612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-171108.log (72acd465746a…), FinalLightCurve_WASP-2 b_2017-11-07.png (9357b00413b9…), FinalLightCurve_WASP-2 b_2017-11-07.csv (9595865115fc…)

Compute resources

Wall time 140.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)